Web Development Basics

Introductory study guide · Prepared as sample educational content for the StudentHub frontend project.

1. How the web works

A browser requests resources from a server using HTTP or HTTPS. HTML describes document structure, CSS controls presentation, and JavaScript adds behaviour. The browser parses these resources and renders a page.

2. Semantic HTML

Use elements according to their meaning: header, nav, main, section, article, and footer. Headings should form a logical outline. Buttons perform actions; links navigate to destinations.

3. CSS foundations

CSS rules select elements and apply declarations. The box model includes content, padding, border, and margin. Flexbox and Grid help create responsive layouts without fixed positioning for every element.

4. Responsive design

Use flexible widths, relative units, and media queries to adapt layouts. The viewport meta tag helps mobile browsers use the device width. Test at narrow and wide sizes, and avoid horizontal overflow.

5. JavaScript in the browser

JavaScript can select elements, listen for events, validate input, and update content. Keep scripts organised and use addEventListener rather than mixing behaviour into HTML attributes.

6. Accessibility and quality

Provide labels for form fields, useful alternative text for meaningful images, visible focus states, sufficient contrast, and keyboard access. Check links, console errors, and page behaviour before sharing a project.

Mini project

Create a one-page profile with a header, navigation, main content, and footer. Add a responsive card grid and a button that updates a status message.